Debojjal Bagchi

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Education

Doctor of Philosophy — 2023–2028

The University of Texas at Austin
Major: Civil Engineering (Transportation Systems)
GPA: 4.00 / 4.00
Advisor: Dr. Stephen D. Boyles

Master of Science in Engineering (Thesis) — 2023–2025

The University of Texas at Austin
Major: Civil Engineering (Transportation Systems)
GPA: 4.00 / 4.00
Thesis: Error bounds for stochastic user equilibrium traffic assignment [**Link**]
Winner: CUTC Milton Pikarsky Memorial Award for outstanding MS thesis in science and technology
Advisor: Dr. Stephen D. Boyles; *Reader:* Dr. Zhaomiao Guo

Bachelor of Science (Research) — 2019–2023

Indian Institute of Science, Bengaluru
Major: Earth & Environmental Science, *Minor:* Mathematics
GPA: 8.7 / 10.0; *Major GPA*: 9.4 / 10.0; **Institute Gold Medal for highest major GPA**
Thesis: Energy-efficient and safe routing for last-mile logistics [**Link**]
Advisor: Dr. Tarun Rambha

Professional Service

- **Committee Member**, Transportation Research Board Standing Committee AT014 on Intermodal Freight and Truck Transportation (2025–2028).
- **Committee Member**, Transportation Research Board Standing Committee AW010 on Ports and Channels (2025).

Awards and Fellowships

International & National Awards

- **Milton Pikarsky Memorial Award**, Best Master's Thesis in Science and Technology, Council of University Transportation Centers, USA (2025).
- **MITACS Globalink Research Internship (GRI) Award**, Fully Funded Summer Research at a Canadian University, MITACS, Canada (2022).
- **Kishore Vaigyanik Protsahan Yojana (KVPY) Fellowship**, Fully Funded Undergraduate Scholarship and Research Grant, Department of Science and Technology, Government of India (2019).
- **Dhirubhai Ambani Scholarship**, Awarded to Top-Ranking Students in India's Class 12 Board Examinations, Reliance Foundation, India (2019).
- **National Talent Search Examination (NTSE) Scholarship**, Awarded through a Two-Stage Nationwide Competitive Examination, National Council of Educational Research and Training, Government of India (2017).

Regional & University-Level Awards

- **TexITE Student Paper Award**, Second Prize, UT Austin (2026).
- **Best Poster Award**, Graduate Symposium, Department of Civil, Architectural, and Environmental Engineering, UT Austin (2026).
- **Professional Development Award**, Graduate School, UT Austin (2025).
- **Graduate School Fellowship**, UT Austin (2023–2027).
- **Institute Gold Medal**, Indian Institute of Science, Awarded for Highest Major GPA in Undergraduate Studies (2023).
- **Telegraph School Award**, Special Honor for Academic Excellence by a Student (2019).
- **Jagadish Bose National Talent Search (JBNSTS) Fellowship** (2018).

Sponsored Research Experience

Graduate Research Assistantship — Jan 2026 – Dec 2026 (Ongoing)

"Develop Multi-Modal Maritime-Rail-Roadway Transportation Model for the Texas Inland and Intercoastal Waterways," sponsored by the Texas Department of Transportation, *PI: Dr. Stephen D. Boyles and Dr. C. Tyler Dick*

- Assisted with proposal authorship.
- Developed a statewide multimodal freight model for Texas.
- Collected freight commodity, volume, and routing data across road, rail, and maritime modes.
- Conducted risk assessments to quantify disaster impacts on the Texas multimodal freight network.
- Tested scenarios to identify travel time variations and critical links.

Graduate Research Assistantship — Sept 2025 – Aug 2027 (Ongoing)

"Evaluating Texas Ports Readiness and Opportunities for Alternative Fuels," sponsored by the Texas Department of Transportation, *PI: Dr. Andrea Gold and Dr. Anna McAuley*

- Investigated the readiness for Texas ports to alternative fuels.
- Conducted stakeholder outreach to collect data on system fuel capabilities.
- Identified gaps and opportunities in port alternative fuel systems.
- Performed market demand analysis for alternate fuel use at Texas ports.
- Developed a tool to assess port readiness for alternative fuels.

Graduate Research Assistantship — Sept 2023 – Dec 2024

"Data-Driven Multimodal Freight Modeling for Waterways and Ports," sponsored by Coastal and Hydraulics Laboratory, US Army Corps of Engineers Research and Development Center, *PI: Dr. Stephen D. Boyles*

- Integrated multimodal data from various sources, including Automatic Identification System vessel tracking data.
- Built a discrete-event simulation of the Port of Houston complex, covering waterway, terminals, and landside operations.
- Developed a queueing model for operating capacity and a differential equation model to identify bottlenecks.
- Simulated hurricane and fog disruption scenarios to assess port performance under capacity constraints.
- **Resulted in five conference presentations and four journal articles (two published, two in preparation).**

MITACS Globalink Research Internship — May 2022 – Aug 2022

"Integrating Waste and Resource Management: Data-Driven Optimization of Urban Mining Logistics," sponsored by MITACS at Université du Québec à Trois-Rivières, *PI: Dr. Amina Lamghari*

- Studied reverse logistics network design under uncertainty for wood industries in Quebec.
- Developed a scenario-based Mixed Integer Linear Program (MILP) formulation for the network design problem.
- Developed an Adaptive Large Neighborhood Search (ALNS) heuristic to improve computational performance.
- Solved the MILP using CPLEX and compared results against the ALNS heuristic in Python.

Undergraduate Research Internship — July 2021 – July 2023

"A Local Search Heuristic for the Bi-criterion Steiner Traveling Salesperson Problem with Time Windows (BSTSPTW)," at the Indian Institute of Science, Bengaluru, India, *PI: Dr. Tarun Rambha*

- Extended the Lin-Kernighan-Helsgaun (LKH) heuristic to BSTSPTW.
- Formulated a scalarization-based MILP for BSTSPTW using a single-commodity flow based formulation.
- Proposed a new local search heuristic for BSTSPTW that outperformed both the LKH extension and the MILP on Amazon delivery routes in Austin, TX.
- **Resulted in two conference presentations and a journal article.**

Publications

Preprints and under review:

[P3] **Bagchi, D.** and Boyles, S. D. (2026). Spectral Analysis of the Logit Mapping and Implications for Stochastic User Equilibrium Algorithms. *Under Review in Transportation Research Part B.* https://doi.org/10.48550/arXiv.2605.21843 [Code]

[P2] Robbennolt, J., **Bagchi, D.**, and Boyles, S. D. (2026). Dynamic Traffic Assignment with Uncertain Demand: A Localized Queue Spillback Approach. *Under Review in Transportmetrica B.*

[P1] **Bagchi, D.**†, Bathgate, K.†, Mitchell, K.N., Asborno, M.I., Kress, M.M., and Boyles, S.D. (2025). Measuring Capacities in Multimodal Maritime Port Systems with Anchorage Queues. *Preprint.* https://doi.org/10.48550/arXiv.2509.22961 **(†Equal contribution)**

Published:

[M3] Bathgate, K., **Bagchi, D.**, Mitchell, K.N., Asborno, M.I., Kress, M.M., and Boyles, S.D. (2026). Anchorage Queue Dynamics due to Fog Channel Closures at the Port of Houston. *Journal of Waterway, Port, Coastal, and Ocean Engineering.* https://doi.org/10.1061/jwped5.wweng-2451

[M2] Bathgate, K., **Bagchi, D.**, Mitchell, K.N., Asborno, M.I., Kress, M.M., and Boyles, S.D. (2026). Characterizing Longitudinal Vessel Arrival and Queue Behavior from AIS at the Port of Houston. *Journal of Waterway, Port, Coastal, and Ocean Engineering.* https://doi.org/10.1061/jwped5.wweng-2441

[M1] Agarwal, P.†, **Bagchi, D.**†, Rambha, T., and Pandey, V. (2025). A Bi-criterion Steiner Traveling Salesperson Problem with Time Windows for Last-Mile Electric Vehicle Logistics. *Computers & Operations Research*. https://doi.org/10.1016/j.cor.2025.107286 [Code] **(†Equal contribution)**

Working Papers

[W2] **Bagchi, D.**†, Bathgate, K.†, Mitchell, K.N., Asborno, M.I., Kress, M.M., and Boyles, S.D. (2026). A Simulation Framework for Evaluating Impacts of Expansion Projects on Multimodal Port System Capacity and Resilience. [Code] **(†Equal contribution)**

[W1] **Bagchi, D.** and Boyles, S. D. (2026). Error Bounds for Stochastic User Equilibrium Traffic Assignment.

Conference Presentations

[C15] **Bagchi, D.** and Boyles, S. D. (2026). Theoretical and Algorithmic Advances in Logit-Based Stochastic User Equilibrium Traffic Assignment. Accepted in *Institute for Operations Research and the Management Sciences Annual Meeting 2026* (**INFORMS**), San Francisco, USA **(Invited session)**.

[C14] **Bagchi, D.**, Bathgate, K., Boyles, S. D., and Asborno, M. I. (2026). A Dynamic Ultimate Capacity Framework for Seaport Systems. *Transportation Research Board 105th Annual Meeting* (**TRB**), Washington, D.C., USA **(Lectern session)**.

[C13] Bathgate, K., **Bagchi, D.**, Boyles, S. D., Mitchell, K. N., Asborno, M. I., and Kress, M. M. (2026). Characterizing Anchorage Queue Dynamics due to Channel Fog Closures at the Port of Houston. *Transportation Research Board 105th Annual Meeting* (**TRB**), Washington, D.C., USA **(Poster session)**.

[C12] Robbennolt, J., **Bagchi, D.**, and Boyles, S. D. (2026). Dynamic Traffic Assignment with Uncertain Demand: A Localized Queue Spillback Approach. *Transportation Research Board 105th Annual Meeting* (**TRB**), Washington, D.C., USA **(Poster session)**.

[C11] Robbennolt, J., **Bagchi, D.**, and Boyles, S. D. (2025). Localized Queue Spillback with Uncertain Demand. *10th International Symposium on Dynamic Traffic Assignment* (**DTA2025**), Salerno, Italy.

[C10] **Bagchi, D.** and Boyles, S. D. (2025). Error Bounds for Stochastic User Equilibrium Traffic Assignment. *12th Triennial Symposium on Transportation Analysis conference* (**TRISTAN XII**), Okinawa, Japan.

[C9] **Bagchi, D.**, Bathgate, K., and Boyles, S. D. (2025). A Framework for Measuring Maritime Port System Capacities with Limited Input Data. *Institute for Operations Research and the Management Sciences Annual Meeting 2025* (**INFORMS**), Atlanta, USA **(Invited session)**.

[C8] Bathgate, K., **Bagchi, D.,** and Boyles, S. D. (2025). Simulation Methods for Measuring Multimodal Maritime Freight System Capacity and Resilience: A Case Study at the Port of Houston. *UT Austin Center for Transportation Research Annual Symposium, 2025*, Austin, USA.

[C7] **Bagchi, D.**, Bathgate, K., and Boyles, S. D. (2025). A Queuing-theory-based Operating Capacity Model for Multimodal Port Operations. *Transportation Research Board 104th Annual Meeting* (**TRB**), Washington, D.C., USA **(Lectern session)**.

[C6] Bathgate, K., **Bagchi, D.**, and Boyles, S. D. (2025). Use of AIS Data to Characterize Vessel Mix in Houston Port Operations for Simulation. *Transportation Research Board 104th Annual Meeting* (**TRB**), Washington, D.C., USA **(Lectern session)**.

[C5] Bathgate, K., **Bagchi, D.**, and Boyles, S. D. (2024). Identifying Capacities in a Multimodal Maritime Freight Network. *Institute for Operations Research and the Management Sciences Annual Meeting 2024* (**INFORMS**), Seattle, USA **(Invited session)**.

[C4] **Bagchi, D.**, and Boyles, S. D. (2024). Error Bounds for Stochastic User Equilibrium Traffic Assignment. *Institute for Operations Research and the Management Sciences Annual Meeting 2024* (**INFORMS**), Seattle, USA **(Invited session)**.

[C3] Bathgate, K., **Bagchi, D.,** and Boyles, S. D. (2024). Data-Driven Modeling for Multimodal Port Resilience Assessment. *UT Austin Center for Transportation Research Annual Symposium, 2024*, Austin, USA.

[C2] **Bagchi, D.**, Agarwal, P., Rambha, T., and Pandey, V. (2023). A Local Search Heuristic for Bi-criterion Steiner Traveling Salesperson Problem. *Transportation Research Board 102nd Annual Meeting 2023* (**TRB**), Washington, D.C., USA.

[C1] **Bagchi, D.**, Agarwal, P., Rambha, T., and Pandey, V. (2022). A Local Search Heuristic for Bi-criterion Steiner Traveling Salesperson Problem. *Institute for Operations Research and the Management Sciences Annual Meeting 2022* (**INFORMS**), Indianapolis, USA.

Open Source Projects

Logit-SUE [Code] **Developer**

- Among the fastest open-source Python solvers (as of 2026) for path-based stochastic user equilibrium traffic assignment under the logit route-choice model.
- Implements Newton-GMRES method with superlinear convergence, Barzilai-Borwein rules, and Method of Successive Averages variants with adaptive constant step sizes.
- Benchmarked on large-scale networks from the Transportation Network Test Problems, including Chicago Regional and Philadelphia.

Multimodal Analysis Port Simulation (MAPS) [Code] [Technical manual] **Co-developer**

- Modular discrete-event simulation framework in Python for multimodal port operations across container, liquid bulk, and dry bulk terminals, waterway channel and landside modes including rail and roadways.
- Supports planning, bottleneck detection, and what-if scenario analysis using real-world data inputs, with sample port data included.
- Models capacity and resilience under operational scenarios such as channel expansion, hurricanes, and fog.

Last-Mile EV Logistics [Code] **Co-developer**

- Codebase for safe and energy-efficient routing of last-mile electric freight vehicles, tested on Amazon delivery routes in Austin, TX (data included).
- Solves the bi-criterion Steiner Traveling Salesperson Problem with turn penalties using state-of-the-art heuristics.
- Also includes exact approaches via a mixed integer linear program solved with CPLEX.

Test Scores

- Graduate Record Examination (GRE): **167/170 in Quantitative Reasoning** and 155/170 in Verbal Reasoning, 2023.
- Test of English as a Foreign Language (TOEFL iBT): 109/120, 2023.
- Joint Entrance Examination (Main): **99.3rd percentile among over 1.14 million candidates nationwide**, 2019.
- Joint Entrance Examination (Advanced): Ranked in the **top 7.5% among approximately 161,000 candidates** nationwide and received an offer of admission from the **Indian Institute of Technology Kharagpur**, 2019.
- All India Senior School Certificate Examination, Grade 12: 96%, 2019.
- All India Secondary School Examination, Grade 10: CGPA 10/10, 2017.

Teaching Experience

- **Teaching Assistant**, CE 311S Probability and Statistics for Civil Engineers, University of Texas at Austin (*Fall 2025; Consolidated student feedback rating: 4.22 / 5.00*)
  - Led one weekly discussion and problem-solving section of 20 students for one hour.
  - Conducted one weekly one-hour office hour session.
  - Prepared assignment solutions and assisted with grading.
- **Teaching Assistant**, CE 311S Probability and Statistics for Civil Engineers, University of Texas at Austin (*Spring 2025; Consolidated student feedback rating: 4.02 / 5.00*)
  - Assisted with a major course redesign.
  - Led two weekly one-hour discussion and problem-solving sections of 20 students each.
  - Conducted two weekly one-hour office hour sessions.
  - Prepared assignment solutions and assisted with grading.
- **Engineering Teaching Assistant Certificate**, Cockrell School of Engineering, University of Texas at Austin (2025).

Skills and Interests

- **Research interests:** Freight and maritime logistics, traffic assignment, congestion pricing.
- **Technical skills:** Optimization, discrete-event simulation, machine learning, data analysis, mathematical programming, Markov decision processes, reinforcement learning.
- **Programming languages:** Python (primary), C, Julia, GAUSS, SQL.
- **Coursework:**
  - **Transportation:** Transportation network analysis, dynamic traffic assignment, railway project design & construction, public transportation, traffic engineering, discrete choice modeling, infrastructure systems.
  - **Operations research and mathematics:** Linear programming, non-linear programming, optimization, applied stochastic processes, Markov decision processes, game theory, introduction to computing for AI & ML, linear algebra, real analysis.

Peer Review and Society Service

Journal Referee

- Transportation Research Part B (2 reviews).
- Transportation Research Record (1 review).

Conference Referee

- Transportation Research Board Annual Meeting (8 reviews).

Society Service

- **Community Service Chair**, UT Student Transportation Council, UT Austin chapter of the Institute of Transportation Engineers (2023–2025).
- **Member**, Notebook Drive, Indian Institute of Science (2019–2023).